Ejemplo: considere el siguiente problema de optimización

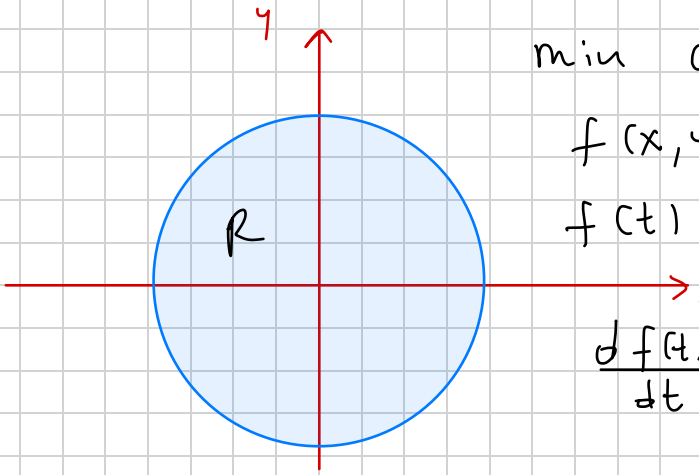
$$\min \quad x+y \quad ; \quad x, y \in R$$

$$R = \{ (x,y) \mid x^2 + y^2 = 1 \}$$

i) Usando una parametrización $x = \cos(t)$, $y = \operatorname{sen}(t)$ $t \in [0, 2\pi]$

ii) Usando multiplicadores de Lagrange.

i)



$$\min \quad \cos(t) + \operatorname{Sen}(t)$$

$$f(x,y) = x + y$$

$$f(t) = \cos(t) + \operatorname{Sen}(t)$$

$$\frac{df(t)}{dt} = -\operatorname{Sen}(t) + \cos(t) = 0$$

$$\operatorname{Sen}(t) = \cos(t)$$

$$\tan(t) = 1$$

$$t = \tan^{-1}(1)$$

$$t = \frac{\pi}{4} \; ; \; \frac{5\pi}{4}$$

t	x	y	$f(x,y)$	
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$\sqrt{2}$	max
$5\pi/4$	$-\sqrt{2}/2$	$-\sqrt{2}/2$	$-\sqrt{2}$	min

ii) Multiplicadores de Lagrange

$$\mathcal{L}(x, y, \lambda) = \underbrace{x + y}_{f(x,y)} + \lambda(x^2 + y^2 - 1)$$

$$\nabla f = \lambda \nabla g$$

$$(1, 1) = \lambda(2x, 2y)$$

$$1 = 2\lambda x \qquad \frac{1}{2\lambda} = x$$

$$1 = 2\lambda y \qquad \frac{1}{2\lambda} = y$$

$$x^2 + y^2 = 1$$

$$\frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 1 \quad * 4\lambda^2$$

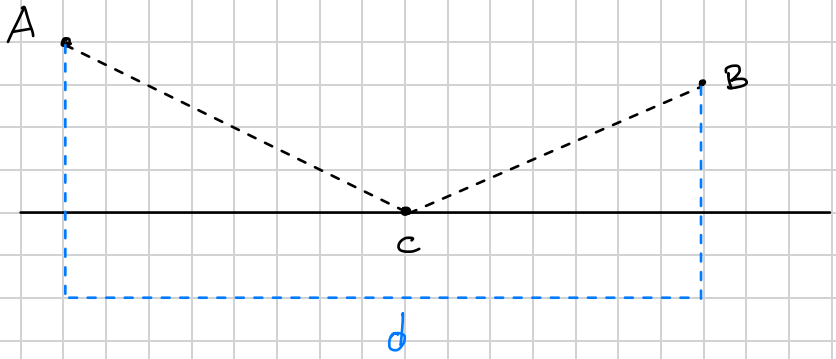
$$2 = 4\lambda^2$$

$$\lambda^2 = \frac{1}{2}$$

$$\lambda = \pm \frac{1}{\sqrt{2}} \; ; \lambda = \pm \frac{\sqrt{2}}{2}$$

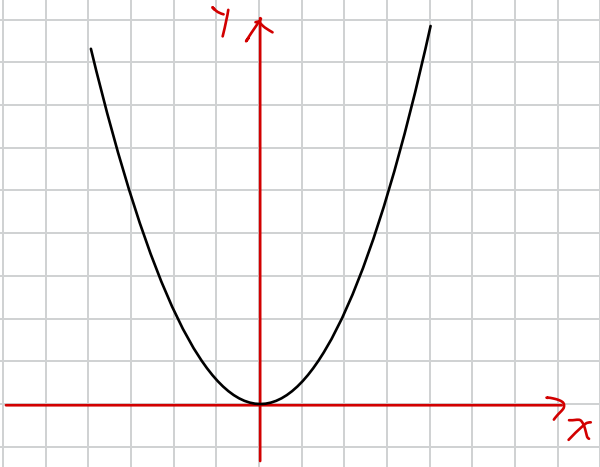
λ	x	y	$f(x,y)$	
$\sqrt{2}/2$	$\sqrt{2}/2$	$\sqrt{2}/2$	$\sqrt{2}$	max
$-\sqrt{2}/2$	$-\sqrt{2}/2$	$-\sqrt{2}/2$	$-\sqrt{2}$	min

Ejercicio propuesto: Minimizar la distancia ACB.



$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

$f(x) = x^2$



$$(\lambda x + (1-\lambda)y)^2 \leq \lambda x^2 + (1-\lambda) y^2$$

$$\lambda^2 x^2 + 2\lambda x(1-\lambda)y + (1-\lambda)^2 y^2 \leq \lambda x^2 + (1-\lambda) y^2$$

$$\lambda^2 x^2 - \lambda x^2 + 2\lambda(1-\lambda) xy + (1-\lambda)^2 y^2 - (1-\lambda) y^2 \leq 0$$

$$(\lambda^2 - \lambda) x^2 + \left((1-\lambda)^2 - (1-\lambda)\right) y^2 + 2\lambda(1-\lambda) xy \leq 0$$

$$1 - 2\lambda + \lambda^2 - 1 + \lambda$$

$$\lambda^2 - \lambda$$

$$\lambda(\lambda - 1)$$

$$\lambda(\lambda-1) x^2 + \lambda(\lambda-1) y^2 - 2\lambda(1-\lambda) xy \leq 0$$

$$\lambda(\lambda-1)(x^2 + y^2 - 2xy) \leq 0$$

$$-\lambda(1-\lambda)(x-y)^2 \leq 0$$

$$\lambda(\lambda-1)(x-y)^2 \leq 0 \quad ; \quad 0 \leq \lambda \leq 1.$$

Lo anterior se cumple para $\lambda \in [0, 1]$ por lo tanto $f(x) = x^2$ es convexa.

Ejemplo: $\min \frac{1}{2} x^T A x \quad x \in \mathbb{R}^n$

$A_{n \times n}$

s.a. $x^T x = 1$

Para que $\frac{1}{2} x^T A x$ sea convexa se debe cumplir que $A \geq 0$ semidefinida positiva. $\lambda_i \geq 0$

$$(1-\lambda)x + \lambda y$$

$$((1-\lambda)x + \lambda y)^T M ((1-\lambda)x + \lambda y)$$

$$(1-\lambda) x^T M (1-\lambda) x + \lambda y^T M \lambda y + 2(1-\lambda) x^T M \lambda y$$

Desigualdad de Cauchy

$$x^T A y \leq \sqrt{(x^T A x)(y^T A y)} \leq \|x\|_A \|y\|_A$$

$$(1-\lambda) x^T M \lambda y \leq \sqrt{((1-\lambda) x^T M (1-\lambda) x)(\lambda y^T M \lambda y)}$$

$$(1-\lambda)\lambda \|x\|_M \|y\|_M$$

$$(1-\lambda)^2 x^T M x + \lambda^2 y^T M y + 2(1-\lambda)\lambda \|x\|_M \|y\|_M$$

$$(1-\lambda)^2 \|x\|_M^2 + \lambda^2 \|y\|_M^2 + 2(1-\lambda)\lambda \|x\|_M \|y\|_M$$

$$(1-\lambda)^2 + \lambda^2 + 2(1-\lambda)\lambda$$

$$1 - 2\lambda + \lambda^2 + \lambda^2 + 2\lambda - 2\lambda^2$$

$$1 \in x^T x = 1.$$